LESSViT: Robust Hyperspectral Representation Learning under Spectral Configuration Shift

Haozhe Si[1], Yuxuan Wan[1], Yuqing Wang[1], Minh N. Do[1,2], Han Zhao[1]
[1]University of Illinois Urbana-Champaign, IL, USA
[2]VinUniversity, Hanoi, Vietnam

Abstract

Modeling hyperspectral imagery (HSI) across different sensors presents a fundamental challenge due to variations in wavelength coverage, band sampling, and channel dimensionality. As a result, models trained under a fixed spectral configuration often fail to generalize to other sensors. Existing Vision Transformer (ViT) approaches either rely on implicit spectral modeling with fixed channel assumptions or adopt explicit spatial–spectral attention with prohibitive computational cost, leading to a fundamental trade-off between efficiency and expressiveness. In this work, we introduce Low-rank Efficient Spatial–Spectral ViT (LESSViT), a sensor-flexible architecture for cross-spectral generalization. LESSViT is built on LESS Attention, a structured low-rank factorization that models joint spatial–spectral interactions through separable spatial and spectral components, reducing the complexity of full spatial–spectral attention from $\mathcal{O}(N^2C^2)$ to $\mathcal{O}(rNC)$, where N is the number of spatial tokens, C is the number of spectral channels, and r is the rank of the low-rank approximation. We further incorporate channel-agnostic patch embedding and wavelength-aware positional encoding to support flexible spectral inputs. To enable efficient and robust pretraining, we introduce a hyperspectral masked autoencoder (HyperMAE) with decoupled spatial–spectral masking and hierarchical channel sampling. We evaluate LESSViT under a cross-spectral generalization setting that simulates cross-sensor variability. Experiments on the SpectralEarth benchmark demonstrate that LESSViT improves robustness under spectral shifts while remaining competitive in-distribution, and explicit and efficient spatial–spectral modeling is essential for scalable and generalizable hyperspectral representation learning. The code is available through the project page: `https://uiuctml.github.io/LESSViT/`.

1 Introduction

Hyperspectral imagery (HSI) captures fine-grained spectral signatures across hundreds of contiguous wavelength bands, enabling detailed characterization of material composition [Zheng et al., 2018, Huang et al., 2022, Lu et al., 2014, Hege et al., 2004], biophysical states [Lu et al., 2020], and environmental processes [Calin et al., 2021, Adão et al., 2017]. Unlike natural images, where channels correspond to broad RGB responses, hyperspectral channels represent physically grounded measurements indexed by wavelength. In practice, different platforms exhibit substantial differences in wavelength coverage, band sampling, and channel dimensionality [Guanter et al., 2015, Pearlman et al., 2003, Behmann et al., 2018]. Consequently, the same scene can produce sensor-dependent spectral responses, introducing a spectral shift that challenges generalization across configurations.

A key challenge in hyperspectral modeling is achieving generalization across varying spectral configurations. Unlike natural image domains with fixed RGB channels, hyperspectral imagery exhibits substantial variation across sensors in wavelength coverage, band sampling, and channel

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

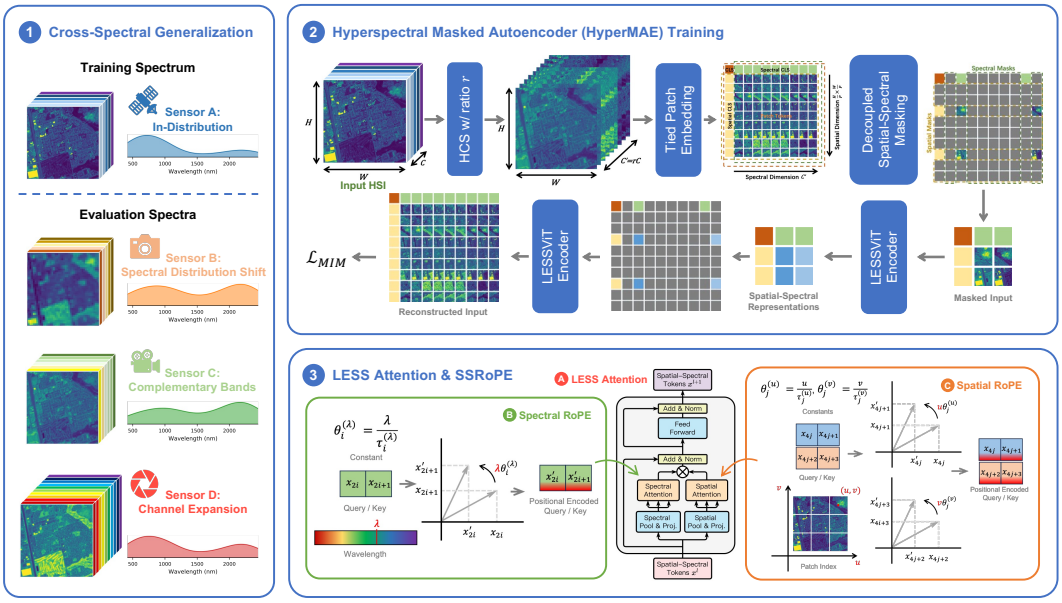


Figure 1: **Overview of LESSViT for cross-spectral generalization.** ①: Cross-spectral generalization setting: train on a fixed spectral configuration and evaluate across spectral configurations with varying wavelength coverage and channel configurations. ②: HyperMAE pretraining with decoupled spatial–spectral masking and hierarchical channel sampling for scalable and robust learning. ③: LESS Attention with SSRoPE for efficient spatial–spectral modeling.

dimensionality, causing models trained under a fixed spectral configuration to fail under varying channel configurations. A natural approach is to explicitly model joint spatial–spectral structure, where spatial variation captures material distribution and spectral variation encodes wavelength-dependent responses. For example, crop-type recognition requires jointly interpreting field structure and vegetation spectra [Zhao et al., 2020], while flood mapping combines spatial context with water-specific spectral absorption [Mateo-Garcia et al., 2021]. However, such explicit modeling in Vision Transformers [Dosovitskiy et al., 2020] incurs $\mathcal{O}(N^2C^2)$ complexity when treating channels as independent tokens, where N is the number of spatial tokens and C is the number of spectral channels, making it computationally prohibitive for hyperspectral inputs. This leads to a fundamental trade-off between efficient but implicit spectral modeling and expressive but computationally expensive spatial–spectral attention.

Existing approaches fail to fully resolve this trade-off. In practice, Vision Transformers are often adapted to hyperspectral data by collapsing the spectral dimension into the channel dimension [Braham et al., 2024, Cong et al., 2022], where cross-channel interactions are only implicitly modeled within the initial patch embedding layer. While computationally efficient, this design assumes a fixed channel structure and treats spectral channels as unordered features, limiting its ability to capture wavelength-dependent relationships and to generalize across varying spectral configurations. DOFA [Xiong et al., 2024] partially alleviates this limitation by introducing wavelength-conditioned patch embeddings, but does not explicitly model spatial–spectral interactions beyond the embedding stage. A more expressive alternative is to treat spectral channels as independent tokens and apply full attention over the resulting spatial–spectral grid [Bao et al., 2023]. This enables explicit spatial–spectral modeling, but incurs $\mathcal{O}(N^2C^2)$ complexity, making it computationally prohibitive for hyperspectral inputs with hundreds of channels. HyperSigma [Wang et al., 2025] reduces this cost via decoupled spatial and spectral branches, but limits direct interaction between them, restricting its ability to capture fine-grained joint dependencies.

To address this fundamental trade-off, we introduce LESS Attention, a structured low-rank factorization of spatial–spectral attention. We exploit the structured interaction between spatial and spectral variations to decompose joint interactions into spatial-only and spectral-only components, while preserving their coupling through a low-rank tensor decomposition. This enables explicit modeling of

joint spatial–spectral dependencies without computing full attention over the entire spatial–spectral grid, reducing complexity from $\mathcal{O}(N^2C^2)$ to $\mathcal{O}(rNC)$, where r controls the approximation rank.

Building on LESS Attention, we develop Low-rank Efficient Spatial–Spectral Vision Transformer (LESSViT), a scalable architecture for cross-spectral generalization. LESSViT employs a tied patch embedding for channel-agnostic tokenization and extends RoPE [Su et al., 2023] to jointly encode spatial coordinates and continuous wavelength information. Together, these designs ensure consistent spatial–spectral structure under varying spectral configurations. To enable scalable training of LESSViT, we adopt a hyperspectral masked autoencoder (HyperMAE) paradigm with decoupled spatial–spectral masking and hierarchical channel sampling, making pretraining both tractable under high channel dimensionality and robust to varying spectral configurations. Together, these designs enable explicit, efficient, and flexible spatial–spectral modeling across varying spectral regimes, as summarized in Figure 1.

We evaluate LESSViT on the SpectralEarth benchmark under a cross-spectral generalization setting that simulates cross-sensor spectral variation. Models are trained on a fixed channel set and evaluated across varying spectral configurations to reflect realistic sensor variability. Results show that LESSViT consistently improves robustness under spectral shifts while remaining competitive under in-distribution settings and scaling effectively to high channel counts. These findings highlight that explicit and efficient spatial–spectral modeling is essential for generalizable hyperspectral representation learning. Our contributions are threefold:

- We propose LESS Attention, a structured low-rank factorization of spatial–spectral attention that reduces the complexity of modeling joint spatial–spectral interactions.
- We develop LESSViT, a channel-agnostic spatial–spectral Vision Transformer with wavelength-aware positional encoding, enabling consistent and robust modeling under varying spectral configurations.
- We establish a cross-spectral generalization evaluation protocol on the SpectralEarth benchmark to systematically assess cross-spectral generalization, demonstrating the effectiveness of explicit and scalable spatial–spectral modeling.

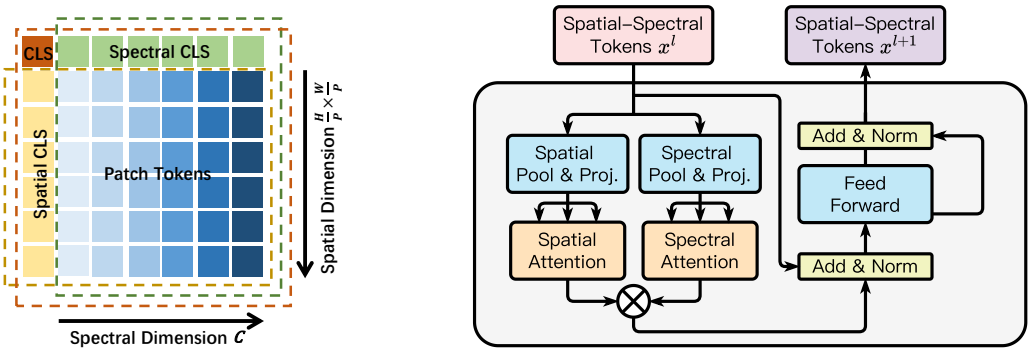
2 Low-rank Efficient Spatial-Spectral ViT (LESSViT)

To achieve generalization across spectral configurations, a model must satisfy three key requirements: *i)* Channel-agnostic tokenization, to support varying spectral dimensionality; *ii)* Explicit spatial–spectral modeling, to capture wavelength-dependent structure; and *iii)* Computational scalability, to remain tractable for high-dimensional inputs. Existing architectures fail to satisfy these requirements simultaneously. To address these challenges, we propose LESSViT, a spatial–spectral ViT that integrates structured tokenization and efficient attention to enable scalable and robust modeling under varying spectral configurations.

2.1 Tied Patch Embedding

A key requirement for cross-spectral generalization is that tokenization should not depend on a fixed channel structure. Standard patch embedding layers assume a fixed number of input channels, making them incompatible with varying spectral configurations. To address this limitation, we employ a *tied patch embedding* layer [Bao et al., 2023] with shared projection weights across channels. Given an input hyperspectral image of size $C \times H \times W$, we partition it into $C \times \frac{H}{P} \times \frac{W}{P}$ patches of size $P \times P$. Each patch is projected into a D-dimensional token using a shared linear mapping $W \in \mathbb{R}^{P^2 \times D}$. This weight-sharing mechanism [Ghiasi et al., 2022] ensures channel-agnostic processing while preserving per-channel structure, enabling the model to operate across varying spectral dimensionalities. The resulting tokens form a spatial–spectral grid of size $\mathbb{R}^{N \times C \times D}$, where $N = \frac{H}{P} \times \frac{W}{P}$. This representation retains the structured organization of hyperspectral data and serves as the foundation for subsequent spatial–spectral modeling.

As illustrated in Figure 2a, we incorporate learnable spatial, spectral, and global `[CLS]` tokens to facilitate structured aggregation. Spatial `[CLS]` tokens summarize information along the spectral dimension, spectral `[CLS]` tokens summarize information along the spatial dimension, and a global



(a) **Spatial-Spectral Tokenization.** (b) **LESS Attention Block.**

Figure 2: **Overview of LESSViT.** (a) The tied patch embedding converts a hyperspectral image ($C \times H \times W$) into a grid of spatial–spectral tokens with spatial, spectral, and global `[CLS]` tokens. (b) The LESS block factorizes spatial and spectral attention via structured decomposition, enabling efficient modeling of joint spatial–spectral interactions.

`[CLS]` token captures joint spatial–spectral information. This yields an augmented token grid of size $\mathbb{R}^{(N+1)\times(C+1)\times D}$, providing structured summaries for the spatial-spectral token grids.

2.2 Low-rank Efficient Spatial-Spectral Attention

To model interactions over spatial–spectral tokens $X \in \mathbb{R}^{N\times C\times D}$, a direct approach flattens tokens and applies standard self-attention [Bao et al., 2023]. However, computing self-attention for NC tokens incurs $\mathcal{O}(N^2C^2D)$ complexity, which is prohibitive for hyperspectral inputs. Consequently, explicit spatial–spectral modeling in this form becomes computationally intractable, failing to meet the scalability requirement. To address this limitation, we exploit the structural separability of hyperspectral data: spatial and spectral variations capture complementary factors and admit a low-rank decomposition. This enables joint spatial–spectral interactions to be modeled through separable components while preserving their coupling.

Based on this observation, we propose the Low-rank Efficient Spatial-Spectral (LESS) attention block, which models joint spatial–spectral attention through a structured low-rank decomposition. Specifically, LESS Attention decomposes interactions into spatial-only and spectral-only components, whose coupling is captured through a low-rank composition. For simplicity, the following formulation considers a single attention head. As shown in Algorithm 1, our LESS attention block first decomposes input tokens X into spatial tokens X_S and spectral tokens X_C using an Attention Pooling Layer (AttenPool). AttenPool aggregates tokens along a selected dimension by using the corresponding `[CLS]` token as a query, producing a weighted summary of all tokens that is then linearly projected to the target sub-dimension.

We then compute spatial attention A_S and spectral attention A_C independently using X_S and X_C. Their Kronecker product defines a structured low-rank form of joint spatial–spectral attention:

$$A := \sum_{i=1}^{r} A_C^i \otimes A_S^i. \tag{1}$$

Accordingly, the attention is given by

$$Y := \sum_{i=1}^{r} (A_C^i \otimes A_S^i)(V_C \otimes V_S) = \sum_{i=1}^{r} (A_C^i V_C) \otimes (A_S^i V_S) = \sum_{i=1}^{r} Y_C^i \otimes Y_S^i, \tag{2}$$

where $V_S \in \mathbb{R}^{N\times d_s}$ and $V_C \in \mathbb{R}^{C\times d_c}$ are the spatial and spectral value representations, respectively, and $Y_S^i \in \mathbb{R}^{N\times d_s}$ and $Y_C^i \in \mathbb{R}^{C\times d_c}$ are the corresponding branch outputs. This formulation avoids constructing the full $NC \times NC$ attention matrix and instead computes interactions through separable spatial and spectral components.

Algorithm 1 Low-Rank Efficient Spatial-Spectral (LESS) Attention Block

Require: Input tokens $X \in \mathbb{R}^{N \times C \times D}$, Sub-dimensions $d_s d_c = D$
Ensure: Output tokens $Y \in \mathbb{R}^{N \times C \times D}$

Decompose spatial-spectral tokens
$X_S \leftarrow \textsc{AttenPool}(X, \text{dim} = 1) \in \mathbb{R}^{N \times d_s}$ ▷ Spatial-only tokens
$X_C \leftarrow \textsc{AttenPool}(X, \text{dim} = 0) \in \mathbb{R}^{C \times d_c}$ ▷ Spectral-only tokens
Compute spatial-only attention
$Q_S, K_S, V_S \leftarrow X_S W_{Q_S}, X_S W_{K_S}, X_S W_{V_S} \in \mathbb{R}^{N \times d_s}$
$Y_S \leftarrow \textsc{Softmax}(Q_S K_S^\top / \sqrt{d_s}) V_S \in \mathbb{R}^{N \times d_s}$ ▷ Complexity $O(N^2 d_s)$
Compute spectral-only attention
$Q_C, K_C, V_C \leftarrow X_C W_{Q_C}, X_C W_{K_C}, X_C W_{V_C} \in \mathbb{R}^{C \times d_c}$
$Y_C \leftarrow \textsc{Softmax}(Q_C K_C^\top / \sqrt{d_c}) V_C \in \mathbb{R}^{C \times d_c}$ ▷ Complexity $O(C^2 d_c)$
Approximate the spatial-spectral attention
$Y \leftarrow Y_C \otimes Y_S \in \mathbb{R}^{NC \times d_s d_c}$ ▷ Complexity $O(NCD)$
$Y \leftarrow \textsc{Reshape}(Y, (N, C, D))$
return Y

The resulting complexity is reduced from $\mathcal{O}(N^2C^2D)$ to $\mathcal{O}(rN^2d_s + rC^2d_c + rNCD)$, where $d_s d_c = D$ and $r \ll \min(N, C)$. Under the regime where spatial and spectral token counts are balanced relative to the sub-dimensions (i.e., $N/C < d_c$ and $C/N < d_s$), the mixed term NCD dominates the overall complexity. As a result, LESS Attention enables practical scaling to hyperspectral inputs with hundreds of channels, where full spatial–spectral attention is otherwise infeasible.

In practice, we allocate a larger sub-dimension d_s to spatial features and a smaller d_c to spectral features, as spectral dependencies often exhibit lower intrinsic dimensionality. This choice preserves the desired complexity regime while maintaining modeling flexibility. Consequently, LESS Attention enables an efficient approximation of explicit spatial–spectral interactions, reconciling expressive joint modeling with computational scalability.

Spatial-Spectral RoPE. To model spectral relationships consistently under varying spectral configurations, positional encoding must be independent of fixed channel indices and instead reflect the physical structure of the spectrum. We introduce spatial–spectral rotary positional embedding (SSRoPE), an axial design that applies standard 2D RoPE [Heo et al., 2024] over spatial coordinates (u, v) and 1D RoPE [Su et al., 2023] over the spectral dimension using wavelengths λ. This yields relative positional encoding across both axes and aligns naturally with the factorized structure of LESS Attention, enabling separate spatial and spectral modeling while preserving their coupling. The detailed formulation is provided in Appendix B.

2.3 Hyperspectral Masked Autoencoder (HyperMAE)

Applying standard masked autoencoding (MAE) to LESSViT is both structurally incompatible and inefficient under high channel dimensionality. We therefore introduce HyperMAE, a tailored pretraining framework for self-supervision.

Decoupled spatial–spectral masking. Standard MAE applies random masking over flattened tokens, implicitly assuming a unified token space. This breaks the axis-aligned structure required by LESS Attention. Inspired by tube masking in VideoMAE [Tong et al., 2022], we apply identical spatial masks across spectral channels rather than temporal frames, and independently mask spectral channels. We use a masking ratio of 75% along each axis. This preserves the separable token grid required by the factorized attention while exposing the model to partial spatial and spectral observations.

Hierarchical Channel Sampling (HCS). While LESS Attention enables efficient spatial–spectral modeling, the HyperMAE decoder for dense reconstruction increases memory and computation, especially at high channel counts. To mitigate this, we adopt hierarchical channel sampling [Bao et al., 2023], where only a subset of channels is randomly sampled at each iteration within a predefined ratio range $[r_l, r_h]$. This reduces reconstruction cost while maintaining exposure to diverse partial-channel inputs. HCS is used only during pretraining. During fine-tuning and inference, the decoder is removed and replaced with a lightweight task head, and computation is dominated by the encoder. This allows efficient processing of full-channel inputs at test time without incurring dense reconstruction cost.

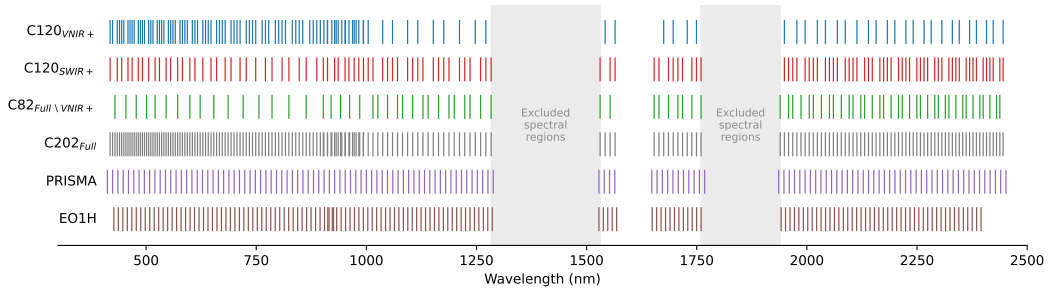


Figure 3: **Band-center locations of the six evaluation configurations.** Each tick marks a retained band center. $C120_{\text{VNIR+}}$ and $C120_{\text{SWIR+}}$ differ in their VNIR–SWIR sampling balance; $C82$ contains the complementary bands interleaved with the training bands, and $C202$ includes all available EnMAP bands. PRISMA and EO1H show the target band centers of the SRF-based configurations. Shaded regions indicate excluded spectral regions.

3 Experiments

We evaluate LESSViT under a cross-spectral generalization setting that captures sensor-induced spectral variability. Models are pretrained and fine-tuned on a fixed spectral configuration and evaluated under different channel configurations designed to isolate distinct forms of spectral shift. We first introduce the cross-spectral generalization task and the experimental protocol. We then present quantitative results and ablation studies to analyze performance, robustness, and efficiency.

3.1 Cross-Spectral Generalization

SpectralEarth [Braham et al., 2024] provides a large-scale hyperspectral pretraining dataset and evaluation benchmark spanning diverse geospatial tasks. In practice, different sensors capture distinct spectral signatures due to variations in wavelength coverage and band sampling. Building on this dataset, we construct a cross-spectral generalization setting by varying channel configurations while keeping spatial content fixed, enabling evaluation that isolates spectral variation.

SpectralEarth provides 202 hyperspectral channels from EnMAP [Guanter et al., 2015]. We partition the spectrum into visible and near-infrared (VNIR) and shortwave infrared (SWIR) regions using a threshold at 1000 nm, resulting in 100 VNIR channels and 102 SWIR channels. Based on this partition, we construct multiple channel subsets that emulate different sensor designs:

Pro-VNIR configuration ($C120_{\text{VNIR+}}$): 120 channels composed of 80 VNIR and 40 SWIR bands, uniformly sampled within each spectral region.

Pro-SWIR configuration ($C120_{\text{SWIR+}}$): 120 channels composed of 40 VNIR and 80 SWIR bands, evaluating robustness to a shift in spectral sampling from VNIR toward SWIR while keeping the channel count fixed.

Complementary configuration ($C82_{\text{Full}\setminus\text{VNIR+}}$): 82 channels comprising 20 VNIR and 62 SWIR bands that complement $C120_{\text{VNIR+}}$ and are spectrally interleaved with its bands, evaluating generalization to unobserved band centers through spectral interpolation.

Full-spectrum configuration ($C202_{\text{Full}}$): all available channels, representing a setting where a model trained with 120 channels must generalize to a larger input channel set without retraining or architectural changes.

SRF-based configurations: We synthesize PRISMA [Vangi et al., 2021] and EO-1 Hyperion [Pearlman et al., 2003] observations by convolving EnMAP spectra with Gaussian spectral response functions (SRFs) defined by each sensor's published band centers and full widths at half maximum (FWHMs). After excluding bands with insufficient spectral support or strong atmospheric absorption, we retain 133 PRISMA-like and 155 Hyperion-like bands. These settings test the model robustness to sensor-dependent spectral mixing beyond channel-subset selection.

The band-center locations of all six configurations are visualized in Figure 3. In our experiments, models are pretrained and fine-tuned on the $C120_{\text{VNIR+}}$ configuration and evaluated under five settings:

Table 1: **Cross-spectral generalization across five downstream tasks.** Scores are means over five seeds; percentages indicate relative drops from the in-distribution configuration. **Bold** denotes the highest score within each dataset–configuration pair and the lowest average relative drop in the summary block.

	CDL (mIoU)						EuroCrops (mIoU)					
Model	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	PRISMA	EO1H	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	PRISMA	EO1H
SpectralViT	**73.33**	$15.11_{\downarrow 79.4\%}$	$12.36_{\downarrow 83.1\%}$	$5.59_{\downarrow 92.4\%}$	$10.84_{\downarrow 85.2\%}$	$7.81_{\downarrow 89.4\%}$	**71.69**	$26.27_{\downarrow 63.4\%}$	$41.14_{\downarrow 42.6\%}$	$46.28_{\downarrow 35.4\%}$	$42.23_{\downarrow 41.1\%}$	$29.32_{\downarrow 59.1\%}$
HyperSigma	44.48	$13.93_{\downarrow 68.7\%}$	$14.50_{\downarrow 67.4\%}$	$14.51_{\downarrow 67.4\%}$	$10.08_{\downarrow 77.3\%}$	$12.86_{\downarrow 71.1\%}$	56.54	$52.16_{\downarrow 7.7\%}$	$48.45_{\downarrow 14.3\%}$	$51.59_{\downarrow 8.8\%}$	$46.32_{\downarrow 18.1\%}$	$38.84_{\downarrow 31.3\%}$
DOFA	40.06	$18.38_{\downarrow 54.1\%}$	$18.08_{\downarrow 54.9\%}$	$12.39_{\downarrow 69.1\%}$	$13.06_{\downarrow 67.4\%}$	$6.39_{\downarrow 84.1\%}$	55.59	$36.85_{\downarrow 33.7\%}$	$32.83_{\downarrow 40.9\%}$	$31.52_{\downarrow 43.3\%}$	$50.28_{\downarrow 9.5\%}$	$50.40_{\downarrow 9.3\%}$
HyperFree	42.50	$28.48_{\downarrow 33.0\%}$	$16.22_{\downarrow 61.8\%}$	$22.16_{\downarrow 47.9\%}$	$28.09_{\downarrow 33.9\%}$	$19.59_{\downarrow 53.9\%}$	52.22	$46.63_{\downarrow 10.7\%}$	$32.87_{\downarrow 37.1\%}$	$36.25_{\downarrow 30.6\%}$	$45.71_{\downarrow 12.5\%}$	$20.91_{\downarrow 60.0\%}$
LESSViT	53.44	$\mathbf{48.32}_{\downarrow 9.6\%}$	$\mathbf{42.08}_{\downarrow 21.3\%}$	$\mathbf{50.75}_{\downarrow 5.0\%}$	$\mathbf{48.96}_{\downarrow 8.4\%}$	$\mathbf{49.05}_{\downarrow 8.2\%}$	57.91	$\mathbf{55.69}_{\downarrow 3.8\%}$	$\mathbf{55.47}_{\downarrow 4.2\%}$	$\mathbf{57.81}_{\downarrow 0.2\%}$	$\mathbf{57.63}_{\downarrow 0.5\%}$	$\mathbf{57.48}_{\downarrow 0.7\%}$

	BDFORET (mIoU)						BNETD (mIoU)					
Model	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	PRISMA	EO1H	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	PRISMA	EO1H
SpectralViT	**77.17**	$48.31_{\downarrow 37.4\%}$	$48.68_{\downarrow 36.9\%}$	$48.73_{\downarrow 36.9\%}$	$46.77_{\downarrow 39.4\%}$	$44.55_{\downarrow 42.3\%}$	**44.69**	$16.86_{\downarrow 62.3\%}$	$17.06_{\downarrow 61.8\%}$	$16.66_{\downarrow 62.7\%}$	$18.00_{\downarrow 59.7\%}$	$16.07_{\downarrow 64.1\%}$
HyperSigma	63.76	$47.97_{\downarrow 24.8\%}$	$44.15_{\downarrow 30.8\%}$	$48.94_{\downarrow 23.2\%}$	$43.68_{\downarrow 31.5\%}$	$9.54_{\downarrow 85.0\%}$	30.36	$18.78_{\downarrow 38.1\%}$	$17.64_{\downarrow 41.9\%}$	$17.68_{\downarrow 41.8\%}$	$15.31_{\downarrow 49.6\%}$	$7.22_{\downarrow 76.2\%}$
DOFA	58.14	$51.26_{\downarrow 11.8\%}$	$42.03_{\downarrow 27.7\%}$	$50.45_{\downarrow 13.2\%}$	$45.04_{\downarrow 22.5\%}$	$2.36_{\downarrow 95.9\%}$	28.88	$18.06_{\downarrow 37.5\%}$	$18.87_{\downarrow 34.7\%}$	$18.89_{\downarrow 34.6\%}$	$15.75_{\downarrow 45.5\%}$	$6.29_{\downarrow 78.2\%}$
HyperFree	58.42	$54.43_{\downarrow 6.8\%}$	$52.63_{\downarrow 9.9\%}$	$54.13_{\downarrow 7.3\%}$	$54.44_{\downarrow 6.8\%}$	$53.93_{\downarrow 7.7\%}$	29.53	$18.53_{\downarrow 37.3\%}$	$16.87_{\downarrow 42.9\%}$	$21.47_{\downarrow 27.3\%}$	$21.93_{\downarrow 25.7\%}$	$18.97_{\downarrow 35.8\%}$
LESSViT	63.87	$\mathbf{62.43}_{\downarrow 2.3\%}$	$\mathbf{60.58}_{\downarrow 5.2\%}$	$\mathbf{63.35}_{\downarrow 0.8\%}$	$\mathbf{62.03}_{\downarrow 2.9\%}$	$\mathbf{62.60}_{\downarrow 2.0\%}$	32.37	$\mathbf{30.76}_{\downarrow 5.0\%}$	$\mathbf{26.35}_{\downarrow 18.6\%}$	$\mathbf{31.25}_{\downarrow 3.5\%}$	$\mathbf{31.80}_{\downarrow 1.8\%}$	$\mathbf{29.62}_{\downarrow 8.5\%}$

	CORINE (mAP)						Avg. Rel. Drop (%)				
Model	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	PRISMA	EO1H	$C120_{\text{SW+}}$	$C82$	$C202$	PRISMA	EO1H
SpectralViT	**78.93**	$36.90_{\downarrow 53.2\%}$	$36.92_{\downarrow 53.2\%}$	$38.37_{\downarrow 51.4\%}$	$35.24_{\downarrow 55.4\%}$	$33.08_{\downarrow 58.1\%}$	59.1	55.5	55.8	56.2	62.6
HyperSigma	69.21	$36.05_{\downarrow 47.9\%}$	$27.56_{\downarrow 60.2\%}$	$47.11_{\downarrow 31.9\%}$	$38.60_{\downarrow 44.2\%}$	$39.48_{\downarrow 43.0\%}$	37.4	42.9	34.6	44.1	61.3
DOFA	66.40	$37.89_{\downarrow 42.9\%}$	$48.27_{\downarrow 27.3\%}$	$35.33_{\downarrow 46.8\%}$	$25.66_{\downarrow 61.4\%}$	$17.58_{\downarrow 73.5\%}$	36.0	37.1	41.4	41.3	68.2
HyperFree	74.04	$47.93_{\downarrow 35.3\%}$	$63.31_{\downarrow 14.5\%}$	$68.48_{\downarrow 7.5\%}$	$65.83_{\downarrow 11.1\%}$	$62.28_{\downarrow 15.9\%}$	24.6	33.2	24.1	18.0	34.6
LESSViT	76.35	$\mathbf{74.21}_{\downarrow 2.8\%}$	$\mathbf{72.95}_{\downarrow 4.5\%}$	$\mathbf{75.99}_{\downarrow 0.5\%}$	$\mathbf{75.05}_{\downarrow 1.7\%}$	$\mathbf{75.01}_{\downarrow 1.8\%}$	**4.7**	**10.7**	**2.0**	**3.0**	**4.2**

(i) *In-distribution* ($C120_{\text{VNIR+}}$), (ii) *Spectral sampling shift* ($C120_{\text{SWIR+}}$), (iii) *Complementary bands* ($C82_{\text{Full}\setminus\text{VNIR+}}$), (iv) *Channel expansion* ($C202_{\text{Full}}$), (v) *SRF shift* (PRISMA and EO1H).

3.2 Experimental Protocol

LESSViT Configuration. We adopt a ViT-Base-sized architecture with 12 encoder and 8 decoder LESS attention blocks, using embedding dimensions of 768 and 512, respectively. Each block employs 12 attention heads. The LESS attention uses rank $r = 1$ with attention ratio $\frac{d_s}{d_c} = 16$, where d_s and d_c denote the per-head spatial and spectral sub-dimensions. The patch size is set to 16. LESSViT is pretrained on the $C120_{\text{VNIR+}}$ configuration of SpectralEarth using the HyperMAE framework for 200 epochs. Additional details are provided in Appendix C.

Baselines. We compare LESSViT with state-of-the-art ViT-based geospatial representation learning methods, including SpectralViT [Braham et al., 2024], DOFA [Xiong et al., 2024], HyperSigma [Wang et al., 2025], and HyperFree [Li et al., 2025]. The released SpectralViT model is pretrained on the full $C202_{\text{Full}}$ spectrum, which would introduce information leakage. We therefore retrain SpectralViT on $C120_{\text{VNIR+}}$ using the same protocol and data as LESSViT to ensure a fair comparison under varying spectral configurations. In contrast, DOFA, HyperSigma, and HyperFree use pretrained weights from their respective datasets and are not exposed to SpectralEarth channel configurations, so we directly adopt their released models. All compared models are subsequently fine-tuned under an identical protocol on $C120_{\text{VNIR+}}$ for 20 epochs, with hyperparameters selected on a held-out validation set, and evaluated across the six spectral configurations. Additional details are provided in Appendix D.

3.3 Quantitative Results

Table 1 summarizes performance across six spectral configurations after fine-tuning on $C120_{\text{VN+}}$, with no further adaptation at evaluation. We report mIoU for segmentation and mAP for multi-label classification, averaged over five random seeds. Full results with standard deviations are provided in the appendix. For each task, the relative performance drop is computed as $1 - C_{\text{OOD}}/C120_{\text{VN+}}$. The summary block reports the unweighted average of these relative drops across the five tasks for each OOD configuration.

Cross-spectral generalization. As shown in Table 1, LESSViT achieves the highest mean performance across all 25 out-of-distribution task–configuration pairs, covering spectral sampling shifts, complementary bands, channel expansion, and simulated sensor responses. Its average relative performance drops range from 2.0% to 10.7%, compared with 18.0%–34.6% for HyperFree, the strongest baseline by this robustness measure. The complementary-band configuration ($C82$) produces the

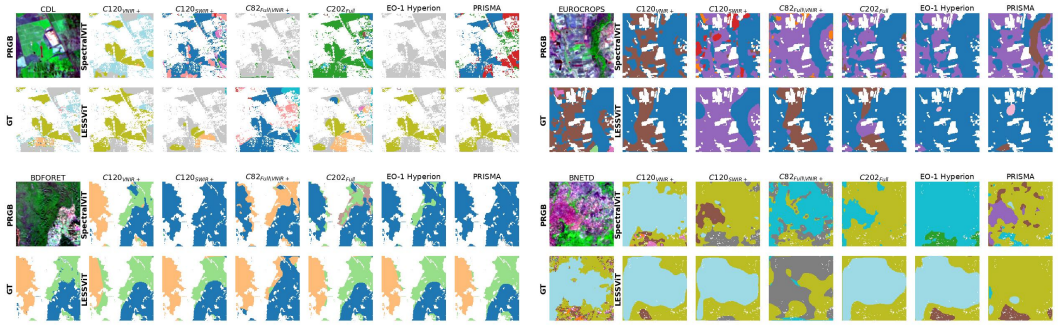


Figure 4: **Qualitative results under cross-spectral generalization.** PRGB denotes pseudo-RGB visualization of hyperspectral inputs, and GT denotes ground-truth segmentation masks. For clearer comparison with GT, background pixels are masked out in the predicted segmentation maps. We show results under different spectral configurations, including in-distribution, spectral shift, complementary bands, channel expansion, and simulated EO-1 Hyperion and PRISMA spectral responses.

largest average drop for LESSViT (10.7%), indicating that generalization to unobserved band centers remains challenging even when they are interleaved with training bands. Under the simulated PRISMA and EO-1 Hyperion responses, LESSViT retains strong performance, with average drops of only 3.0% and 4.2%, respectively. These results demonstrate robustness to both channel-subset reconfiguration and simulated sensor-dependent spectral responses.

In-distribution performance. Under the in-distribution setting ($C120_{\text{VNIR+}}$), SpectralViT consistently achieves the best performance, while LESSViT ranks second in most cases. The strong in-distribution performance suggests that SpectralViT is specialized to the fixed spectral configuration, enabling more effective exploitation of spatial patterns.

Additionally, SpectralViT adopts a smaller patch size (4×4), which preserves finer spatial details and benefits dense prediction tasks. Consistently, the performance gap is smaller on CORINE classification, where spatial resolution is less critical.

Discussion. The observed results reveal a trade-off between in-distribution specialization and cross-spectral generalization. Models such as SpectralViT achieve strong performance under a fixed spectral configuration, but degrade substantially when the channel configuration or spectral response changes. These results suggest that strong in-distribution performance does not necessarily translate into robustness under spectral shifts. In contrast, LESSViT explicitly models spatial–spectral interactions and maintains more stable performance across complementary bands, channel expansion, and simulated SRF-induced spectral mixing. The PRISMA-like and Hyperion-like results extend this robustness beyond channel-subset selection to changes in how spectral measurements are formed. Overall, LESSViT achieves a more favorable balance between in-distribution performance and robustness across the evaluated spectral configurations.

3.4 Qualitative Results

We visualize representative segmentation results under channel-subset reconfiguration in Figure 4, comparing the model with the strongest in-distribution performance (SpectralViT) and the model with the best robustness under varying spectral configurations (LESSViT). In the in-distribution setting, SpectralViT produces more detailed segmentation maps, which can be attributed to its smaller patch size (4×4). This finer spatial resolution has been shown to improve segmentation quality, albeit at the cost of increased token count and computation. Under the cross-spectral generalization setting, both models exhibit performance degradation, with the complementary-band configuration ($C82$) being the most challenging. We further observe that SpectralViT is especially sensitive to spectral shift ($C120_{\text{SWIR+}}$) and channel expansion ($C202$), where it produces large, spatially coherent misclassifications. In contrast, LESSViT maintains more stable predictions under these shifts. Under simulated EO-1 Hyperion and PRISMA responses, LESSViT better preserves the ground-truth spatial patterns in CDL and BDFORET, demonstrating improved robustness to spectral variation. This difference highlights the limitation of relying on a fixed spectral structure in the encoder, which hinders adaptation under changing spectral distributions, a challenge that LESSViT successfully addresses through flexible spatial–spectral modeling.

Table 2: **Comparison of attention operators under spectral configuration shift.** Avg. Seg. denotes the average mIoU across the four segmentation tasks. Bold indicates the best score in each column.

	Avg. Seg. (mIoU)				CORINE (mAP)			
Attention operator	$C120_{\mathrm{VN+}}$	$C120_{\mathrm{SW+}}$	$C82$	$C202$	$C120_{\mathrm{VN+}}$	$C120_{\mathrm{SW+}}$	$C82$	$C202$
Spatial Attention	**56.29**	22.76	16.08	29.25	**76.97**	29.32	26.37	33.51
Full Spatial–Spectral Attention	49.05	**46.70**	**46.49**	**44.87**	71.25	70.17	70.78	69.47
LESS Attention	49.19	46.07	43.97	42.88	73.19	**71.18**	**72.14**	**70.00**

3.5 Effect of Attention Design

To distinguish the effect of attention design from that of pretraining, we compare LESS Attention with full spatial–spectral attention and spatial-only attention under the same HyperMAE pretraining protocol, keeping decoupled masking, hierarchical channel sampling, training data, and optimization settings fixed. All three models use a ViT-S backbone with 64×64 inputs to make full spatial–spectral attention computationally feasible.

As shown in Table 2, spatial-only attention achieves the highest in-distribution scores but degrades substantially under spectral reconfiguration, despite receiving the same spectral exposure during pretraining. Both LESS and full spatial–spectral attention retain substantially stronger performance, supporting the contribution of explicit spatial–spectral modeling to cross-spectral generalization. LESS achieves comparable in-distribution segmentation performance and higher CORINE mAP across all configurations, while full attention performs better on average segmentation mIoU under spectral shifts. These results show that LESS Attention achieves broadly comparable cross-spectral performance to full spatial–spectral attention. We next examine how this performance compares in terms of computational scalability.

3.6 Model Efficiency

We measure the inference latency of LESSViT and ChannelViT [Bao et al., 2023] on 2000 samples while progressively increasing the number of input channels. Latency is normalized to each model's corresponding 10-channel setting, with results shown in Figure 5.

As channel count increases, LESSViT maintains near-constant latency, whereas ChannelViT's latency increases rapidly and becomes out of memory at 200 channels under this benchmark configuration. These results demonstrate more favorable scaling with channel count, complementing the broadly comparable cross-spectral performance observed in Section 3.5. Additional measurements of absolute latency, memory usage, and training cost are provided in Appendix F.5.

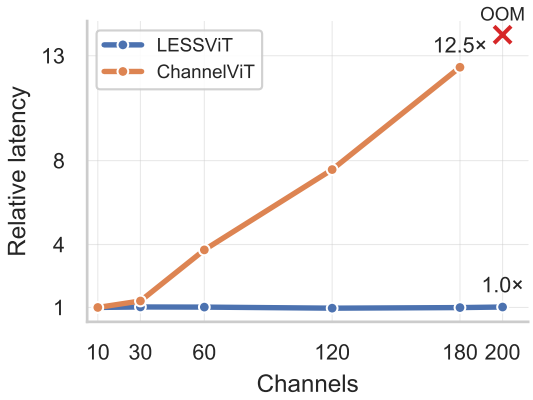


Figure 5: **Inference latency vs. channel count.** Latency is normalized to the C=10 setting. ChannelViT encounters an out-of-memory (OOM) error at $C = 200$ on our hardware (144 GB GPU).

3.7 Impact of Proposed Modules

We conduct ablations to evaluate key components in LESSViT. All models use a ViT-S-sized backbone and are pretrained with HyperMAE for 50 epochs, with the same training and evaluation protocol as the main experiments. Partial results are shown in Table 3, with full results in Appendix F.4. We study (i) positional encoding, comparing SSRoPE with SIREN [Sitzmann et al., 2020], and (ii) hierarchical channel sampling (HCS) with varying retained ratios r_{HCS}.

Positional encoding. Replacing SSRoPE with SIREN degrades performance on segmentation tasks. On in-distribution segmentation tasks, SSRoPE improves spatial prediction quality, suggesting that spatial RoPE provides more effective positional encoding for dense prediction. Under spectral variation, SSRoPE yields more stable performance, indicating that wavelength-aware spectral RoPE supports generalization across varying channel configurations. On CORINE classification, the gap is

Table 3: Ablation of positional encoding and hierarchical channel sampling (HCS) under cross-spectral generalization. Models are trained on $C120_{\text{VN+}}$ and evaluated across spectral configurations. *Avg. Seg.* denotes the average mIoU across all segmentation tasks.

			CDL (mIoU)				CORINE (mAP)				*Avg. Seg.* (mIoU)			
Config.	PosEmb	r_{HCS}	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$
Default	SSRoPE	$[0.2, 0.3]$	40.04	41.22	35.60	40.50	70.74	68.30	67.11	69.61	44.48	42.66	39.96	43.63
w/o SSRoPE	SIREN	$[0.2, 0.3]$	28.79	29.60	30.58	29.82	70.32	69.48	64.68	69.89	40.04	37.76	37.84	38.65
High r_{HCS}	SSRoPE	$[0.4, 0.5]$	40.93	40.87	35.88	39.13	69.68	69.04	64.61	69.83	45.44	44.09	40.70	43.95

smaller, suggesting lower sensitivity to positional encoding. Overall, SSRoPE benefits dense spatial modeling in ID settings and supports cross-spectral robustness under OOD channel configurations.

Hierarchical channel sampling. Increasing the HCS retained ratio from $[0.2, 0.3]$ to $[0.4, 0.5]$ leads to only marginal performance changes across spectral configurations, suggesting that moderate channel sampling does not substantially degrade representation quality. This is likely because HCS primarily reduces the decoder reconstruction space, while the encoder continues to learn from diverse channel subsets through channel masking. At the same time, using the lower r_{HCS} reduces pretraining time from 41 hours to 34 hours, corresponding to a 17% reduction. These results indicate that HCS improves training efficiency with minimal impact on downstream performance.

4 Related Works

Tensor Product Attention (TPA) [Zhang et al., 2025], a concurrent work, introduces a context-aware factorization of attention by constructing query, key, and value tensors through low-rank decompositions. Instead of using fixed projections, TPA dynamically adapts these factorized representations based on the input, enabling expressive attention modeling while maintaining computational efficiency. Although both TPA and our approach lead to similar low-rank formulations of attention, they arise from different motivations and target different objectives. TPA is primarily designed to improve the flexibility and expressiveness of attention through context-aware factorization of query, key, and value representations. In contrast, our method is motivated by the structural separability of spatial and spectral interactions in hyperspectral data, and aims to explicitly model spatial–spectral interactions in a scalable manner. Therefore, while the formulations share similarities, the two approaches are developed under different problem settings and serve distinct purposes.

5 Conclusions

In this work, we identify cross-spectral generalization as a fundamental challenge in hyperspectral modeling and present LESSViT as a scalable solution. By combining low-rank spatial–spectral attention, channel-agnostic tokenization, and wavelength-aware positional encoding, LESSViT enables explicit and efficient modeling across varying spectral configurations. A tailored HyperMAE pretraining strategy further makes training tractable under high channel dimensionality. Experiments demonstrate that LESSViT consistently improves robustness under spectral reconfiguration while remaining competitive under in-distribution settings.

Despite these advances, several limitations remain. First, our evaluation uses spectral configurations derived from EnMAP, including simulated PRISMA and Hyperion responses, and therefore does not establish real cross-sensor generalization. Differences in calibration, atmospheric conditions, spatial resolution, and sensor noise remain untested. Second, improving segmentation quality by reducing patch size introduces a substantial increase in token count. Under high token counts and high channel dimensionality, even the linear-complexity formulation of LESS Attention remains computationally demanding, limiting scalability in such regimes.

Future work will focus on validating generalization on real cross-sensor observations and developing more efficient architectures for handling high-resolution spatial–spectral inputs. In addition, our current evaluation is limited to geospatial hyperspectral data. Extending LESSViT to other hyperspectral domains, such as medical imaging and material analysis, and validating its effectiveness on large-scale datasets in these settings remains an important direction.

Acknowledgments and Disclosure of Funding

This work was supported by the IBM-Illinois Discovery Accelerator Institute through the IBM-IIDAI joint grant, "Foundation Models for Spatio-temporal and Multi-modal Geospatial Data." The authors would also like to express their sincere gratitude to Professor Hendrik F. Hamann of Stony Brook University for his valuable guidance and insightful discussions, which helped inspire and shape this work. Computational resources used in this work were partially provided through NSF NAIRR grant NAIRR240419 and the NVIDIA Academic Grant Program.

References

Telmo Adão, Jonáš Hruška, Luís Pádua, José Bessa, Emanuel Peres, Raul Morais, and Joaquim Joao Sousa. Hyperspectral imaging: A review on uav-based sensors, data processing and applications for agriculture and forestry. *Remote sensing*, 9(11):1110, 2017.

Yujia Bao, Srinivasan Sivanandan, and Theofanis Karaletsos. Channel vision transformers: An image is worth c x 16 x 16 words. *arXiv preprint arXiv:2309.16108*, 2023.

Jan Behmann, Kelvin Acebron, Dzhaner Emin, Simon Bennertz, Shizue Matsubara, Stefan Thomas, David Bohnenkamp, Matheus T. Kuska, Jouni Jussila, Harri Salo, Anne-Katrin Mahlein, and Uwe Rascher. Specim iq: Evaluation of a new, miniaturized handheld hyperspectral camera and its application for plant phenotyping and disease detection. *Sensors*, 18(2), 2018. ISSN 1424-8220. doi: 10.3390/s18020441. URL `https://www.mdpi.com/1424-8220/18/2/441`.

Claire Boryan, Zhengwei Yang, Rick Mueller, and Mike Craig. Monitoring us agriculture: the us department of agriculture, national agricultural statistics service, cropland data layer program. *Geocarto International*, 26(5):341–358, 2011.

Nassim Ait Ali Braham, Conrad M Albrecht, Julien Mairal, Jocelyn Chanussot, Yi Wang, and Xiao Xiang Zhu. Spectralearth: Training hyperspectral foundation models at scale. *arXiv preprint arXiv:2408.08447*, 2024.

Mihaela Antonina Calin, Adrian Cantemir Calin, and Doina Nicoleta Nicolae. Application of airborne and spaceborne hyperspectral imaging techniques for atmospheric research: Past, present, and future. *Applied Spectroscopy Reviews*, 56(4):289–323, 2021.

Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. Longlora: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307*, 2023.

Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David Lobell, and Stefano Ermon. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems*, 35:197–211, 2022.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

European Environment Agency. Corine land cover (clc) 2018, version 2020_20u1. `https://land.copernicus.eu/pan-european/corine-land-cover`, 2020. Copernicus Land Monitoring Service, European Environment Agency.

Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann Lecun. Rankme: Assessing the downstream performance of pretrained self-supervised representations by their rank, 2023. URL `https://arxiv.org/abs/2210.02885`.

Charles Gaydon and Floryne Roche. Pureforest: A large-scale aerial lidar and aerial imagery dataset for tree species classification in monospecific forests. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5895–5904. IEEE, 2025.

Amin Ghiasi, Hamid Kazemi, Eitan Borgnia, Steven Reich, Manli Shu, Micah Goldblum, Andrew Gordon Wilson, and Tom Goldstein. What do vision transformers learn? a visual exploration. *arXiv preprint arXiv:2212.06727*, 2022.

Luis Guanter, Hermann Kaufmann, Karl Segl, Saskia Foerster, Christian Rogass, Sabine Chabrillat, Theres Kuester, André Hollstein, Godela Rossner, Christian Chlebek, Christoph Straif, Sebastian Fischer, Stefanie Schrader, Tobias Storch, Uta Heiden, Andreas Mueller, Martin Bachmann, Helmut Mühle, Rupert Müller, Martin Habermeyer, Andreas Ohndorf, Joachim Hill, Henning Buddenbaum, Patrick Hostert, Sebastian Van der Linden, Pedro J. Leitão, Andreas Rabe, Roland Doerffer, Hajo Krasemann, Hongyan Xi, Wolfram Mauser, Tobias Hank, Matthias Locherer, Michael Rast, Karl Staenz, and Bernhard Sang. The enmap spaceborne imaging spectroscopy mission for earth observation. *Remote Sensing*, 7(7):8830–8857, 2015. ISSN 2072-4292. doi: 10.3390/rs70708830. URL `https://www.mdpi.com/2072-4292/7/7/8830`.

E Keith Hege, Dan O'Connell, William Johnson, Shridhar Basty, and Eustace L Dereniak. Hyperspectral imaging for astronomy and space surveillance. In *Imaging Spectrometry IX*, volume 5159, pages 380–391. SPIE, 2004.

Byeongho Heo, Song Park, Dongyoon Han, and Sangdoo Yun. Rotary position embedding for vision transformer. In *European Conference on Computer Vision*, pages 289–305. Springer, 2024.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3, 2022.

Shuan-Yu Huang, Arvind Mukundan, Yu-Ming Tsao, Youngjo Kim, Fen-Chi Lin, and Hsiang-Chen Wang. Recent advances in counterfeit art, document, photo, hologram, and currency detection using hyperspectral imaging. *Sensors*, 22(19):7308, 2022.

Ting Jiang, Shaohan Huang, Shengyue Luo, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, Qi Zhang, Deqing Wang, et al. Mora: High-rank updating for parameter-efficient fine-tuning. *arXiv preprint arXiv:2405.12130*, 2024.

Weili Kong, Baisen Liu, Xiaojun Bi, Changdong Yu, Xinyao Li, and Yushi Chen. Hypersl: A spectral foundation model for hyperspectral image interpretation. *IEEE Transactions on Geoscience and Remote Sensing*, 63:1–19, 2025.

Jingtao Li, Yingyi Liu, Xinyu Wang, Yunning Peng, Chen Sun, Shaoyu Wang, Zhendong Sun, Tian Ke, Xiao Jiang, Tangwei Lu, Anran Zhao, and Yanfei Zhong. Hyperfree: A channel-adaptive and tuning-free foundation model for hyperspectral remote sensing imagery, 2025. URL `https://arxiv.org/abs/2503.21841`.

Vladislav Lialin, Namrata Shivagunde, Sherin Muckatira, and Anna Rumshisky. Relora: High-rank training through low-rank updates. *arXiv preprint arXiv:2307.05695*, 2023.

Weiwei Liu, Kai Liu, Weiwei Sun, Gang Yang, Kai Ren, Xiangchao Meng, and Jiangtao Peng. Self-supervised feature learning based on spectral masking for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–15, 2023.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

Bing Lu, Phuong D Dao, Jiangui Liu, Yuhong He, and Jiali Shang. Recent advances of hyperspectral imaging technology and applications in agriculture. *Remote sensing*, 12(16):2659, 2020.

Guolan Lu, Luma Halig, Dongsheng Wang, Xulei Qin, Zhuo Georgia Chen, and Baowei Fei. Spectral-spatial classification for noninvasive cancer detection using hyperspectral imaging. *Journal of biomedical optics*, 19(10):106004–106004, 2014.

Gonzalo Mateo-Garcia, Joshua Veitch-Michaelis, Lewis Smith, Silviu Vlad Oprea, Guy Schumann, Yarin Gal, Atılım Güneş Baydin, and Dietmar Backes. Towards global flood mapping onboard low cost satellites with machine learning. *Scientific reports*, 11(1):7249, 2021.

J.S. Pearlman, P.S. Barry, C.C. Segal, J. Shepanski, D. Beiso, and S.L. Carman. Hyperion, a space-based imaging spectrometer. *IEEE Transactions on Geoscience and Remote Sensing*, 41(6): 1160–1173, 2003. doi: 10.1109/TGRS.2003.815018.

Weijieying Ren, Xinlong Li, Lei Wang, Tianxiang Zhao, and Wei Qin. Analyzing and reducing catastrophic forgetting in parameter efficient tuning. *arXiv preprint arXiv:2402.18865*, 2024.

Linus Scheibenreif, Michael Mommert, and Damian Borth. Masked vision transformers for hyperspectral image classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2166–2176, 2023.

Maja Schneider, Amelie Broszeit, and Marco Körner. Eurocrops: A pan-european dataset for time series crop type classification. *arXiv preprint arXiv:2106.08151*, 2021.

Vincent Sitzmann, Julien Martel, Alexander Bergman, David Lindell, and Gordon Wetzstein. Implicit neural representations with periodic activation functions. *Advances in neural information processing systems*, 33:7462–7473, 2020.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding, 2023. URL `https://arxiv.org/abs/2104.09864`.

Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. *Advances in neural information processing systems*, 35:10078–10093, 2022.

Elia Vangi, Giovanni D'Amico, Saverio Francini, Francesca Giannetti, Bruno Lasserre, Marco Marchetti, and Gherardo Chirici. The new hyperspectral satellite prisma: Imagery for forest types discrimination. *Sensors*, 21(4), 2021. ISSN 1424-8220. doi: 10.3390/s21041182. URL `https://www.mdpi.com/1424-8220/21/4/1182`.

Di Wang, Meiqi Hu, Yao Jin, Yuchun Miao, Jiaqi Yang, Yichu Xu, Xiaolei Qin, Jiaqi Ma, Lingyu Sun, Chenxing Li, et al. Hypersigma: Hyperspectral intelligence comprehension foundation model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

Yue Wang, Ming Wen, Hailiang Zhang, Jinyu Sun, Qiong Yang, Zhimin Zhang, and Hongmei Lu. Hsimae: A unified masked autoencoder with large-scale pre-training for hyperspectral image classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024.

Zhitong Xiong, Yi Wang, Fahong Zhang, Adam J Stewart, Joëlle Hanna, Damian Borth, Ioannis Papoutsis, Bertrand Le Saux, Gustau Camps-Valls, and Xiao Xiang Zhu. Neural plasticity-inspired multimodal foundation model for earth observation. *arXiv preprint arXiv:2403.15356*, 2024.

Hengyu Zhang. Sinklora: Enhanced efficiency and chat capabilities for long-context large language models. *arXiv preprint arXiv:2406.05678*, 2024.

Yifan Zhang, Yifeng Liu, Huizhuo Yuan, Zhen Qin, Yang Yuan, Quanquan Gu, and Andrew Chi-Chih Yao. Tensor product attention is all you need. *arXiv preprint arXiv:2501.06425*, 2025.

Hongbo Zhao, Bolin Ni, Junsong Fan, Yuxi Wang, Yuntao Chen, Gaofeng Meng, and Zhaoxiang Zhang. Continual forgetting for pre-trained vision models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28631–28642, 2024.

Ji Zhao, Yanfei Zhong, Xin Hu, Lifei Wei, and Liangpei Zhang. A robust spectral-spatial approach to identifying heterogeneous crops using remote sensing imagery with high spectral and spatial resolutions. *Remote Sensing of Environment*, 239:111605, 2020.

Xiangyu Zhao, Zhitong Xiong, and Xiao Xiang Zhu. Hysens: Sensor-agnostic foundation models for hyperspectral data. *IEEE Transactions on Geoscience and Remote Sensing*, 2026.

Yan Zheng, Jiarui Bai, Jingna Xu, Xiayang Li, and Yimin Zhang. A discrimination model in waste plastics sorting using nir hyperspectral imaging system. *Waste Management*, 72:87–98, 2018.

Lingxuan Zhu, Jiaji Wu, Wang Biao, Yi Liao, and Dandan Gu. Spectralmae: Spectral masked autoencoder for hyperspectral remote sensing image reconstruction. *Sensors*, 23(7):3728, 2023.

A Additional Related Works

Hyperspectral Modeling. Hyperspectral (HSI) and multispectral (MSI) models for geospatial applications adopt different strategies for capturing spatial–spectral interactions. One line of work aggregates spectral channels during token embedding and subsequently applies attention over spatial tokens. SpectralViT [Braham et al., 2024] projects spectral channels into latent features, while SatMAE [Cong et al., 2022] uses channel grouping to retain spectral structure. DOFA [Xiong et al., 2024] introduces wavelength-conditioned input parameterization, and HyperFree [Li et al., 2025] dynamically constructs input embeddings from a learned spectral weight dictionary. Although these designs incorporate spectral information into token embeddings, subsequent attention does not explicitly operate over individual spectral channels.

Another line of work explicitly models spatial–spectral interactions. ChannelViT [Bao et al., 2023] applies attention jointly over spatial and channel tokens, incurring quadratic complexity in the total token count. HyperSigma [Wang et al., 2025] models spatial and spectral dependencies in separate branches and fuses their features. LESSViT preserves explicit joint spatial–spectral modeling while reducing its computational cost through structured low-rank attention.

Complementary work explores spectral pretraining and adaptation to heterogeneous spectral inputs. SpectralMAE [Zhu et al., 2023] reconstructs missing spectral bands, while Liu et al. [2023] use masked spectral reconstruction to learn representations for hyperspectral classification. HyperSL [Kong et al., 2025] combines masked pretraining with unified spectral tokens and wavelength-aware positional encoding, and HySens [Zhao et al., 2026] harmonizes spectra onto a common wavelength grid with wavelength-based embeddings. These studies motivate learning transferable spectral representations.

Hyperspectral Datasets. Recent large-scale pretraining datasets and benchmarks such as Braham et al. [2024], Scheibenreif et al. [2023], Wang et al. [2024, 2025] have significantly expanded the scope of hyperspectral representation learning, introducing data with high spectral dimensionality and varying spatial resolutions. These settings exacerbate the scalability challenges of attention-based models and highlight the need for architectures that can generalize across varying spectral configurations.

Tensor Factorization. Low-rank factorization has been widely adopted to reduce computational cost and parameter complexity in deep learning models. A prominent line of work is the LoRA family [Hu et al., 2022], which decomposes weight updates into low-rank matrices during fine-tuning. Subsequent extensions have explored various training regimes, including efficient pretraining [Lialin et al., 2023, Jiang et al., 2024], long-context modeling [Chen et al., 2023, Zhang, 2024], and continual learning [Zhao et al., 2024, Ren et al., 2024]. These methods primarily operate on model parameters and construct static low-rank approximations that are independent of the input structure.

B Spatial-Spectral RoPE

To encode both spatial information and spectral structure of patches, we introduce spatial–spectral rotary positional embedding (SSRoPE). Our construction is axial: we encode the two spatial coordinates separately and encode the spectral dimension using the physical central wavelength associated with each band. Separate rotary operators are constructed for the spatial and spectral branches and applied within the corresponding branch-specific attention computation.

Let a patch token correspond to spatial position (u, v) and spectral band with central wavelength λ. For each attention head, we partition the spatial branch into two equal parts corresponding to the horizontal and vertical axes. For frequency index i, we define

$$\theta_i^{(u)} = \frac{u}{\tau_i^{(u)}}, \quad \theta_i^{(v)} = \frac{v}{\tau_i^{(v)}}, \quad \theta_i^{(\lambda)} = \frac{\lambda}{\tau_i^{(\lambda)}}, \tag{3}$$

where $\tau_i^{(u)}, \tau_i^{(v)}, \tau_i^{(\lambda)}$ are geometric periods.

These phases define block-wise rotations applied to attention features. Denoting by $R(\theta)$ the standard 2D rotation on paired feature dimensions, we construct spatial and spectral rotary operators as

$$R_{\mathrm{s}} = \mathrm{diag}\Big(R(\theta_1^{(u)}), \ldots, R(\theta_{d_s/4}^{(u)}), R(\theta_1^{(v)}), \ldots, R(\theta_{d_s/4}^{(v)})\Big), \tag{4}$$

$$R_{\mathrm{c}} = \mathrm{diag}\Big(R(\theta_1^{(\lambda)}), \ldots, R(\theta_{d_c/2}^{(\lambda)})\Big), \tag{5}$$

where d_s and d_c denote the spatial and spectral branch dimensions within each attention head, following the single-head notation in Section 2.2.

We then apply the rotary transformation to the query and key tensors in each branch:

$$\tilde{Q}_s(u,v) = R_{\mathrm{s}}(u,v)Q_s(u,v), \quad \tilde{K}_s(u,v) = R_{\mathrm{s}}(u,v)K_s(u,v), \tag{6}$$

$$\tilde{Q}_c(\lambda) = R_{\mathrm{c}}(\lambda)Q_c(\lambda), \quad \tilde{K}_c(\lambda) = R_{\mathrm{c}}(\lambda)K_c(\lambda). \tag{7}$$

Following standard RoPE [Su et al., 2023], the transformation is applied only to non-`[CLS]` tokens.

C Pretraining Details

C.1 Data

We pretrain our model using the $C120_{\text{VNIR+}}$ spectral configuration of the SpectralEarth dataset [Braham et al., 2024], which is a large-scale hyperspectral Earth Observation dataset built from the EnMAP [Guanter et al., 2015] satellite. Each HSI includes 202 channels. The dataset covers 538k locations worldwide. The spatial resolution of the images is 3.84km $\times$ 3.84km. For further details about the dataset, please refer to Braham et al. [2024].

C.2 Implementation

We train our main model for 200 epochs on 4$\times$NVIDIA H200 GPUs, each with 144G VRAM, which takes 96 hours. For comparison, pretraining SpectralViT-B on the same hardware takes 28 hours. LESSViT is trained with an effective batch size of 1024, bfloat16 precision, a base learning rate of 1.5e-4, warmup for 5% of the total epochs, and cooldown via a cosine annealing scheduler. We optimize the model using the AdamW Loshchilov and Hutter [2017] optimizer with a weight decay of 5e-2. We normalize data following Braham et al. [2024]. For data augmentation, we apply the random horizontal flip with a probability of 50%. We do not resize the image to preserve the physical property of spatial resolution within the geospatial data.

D Evaluation Details

We evaluate our pretrained LESSViT models and other baseline models on released SpectralEarth benchmarks: CDL [Boryan et al., 2011], EuroCrops [Schneider et al., 2021], CORINE [European Environment Agency, 2020], BDFORET [Gaydon and Roche, 2025] and BNETD [Braham et al., 2024]. Following SpectralEarth's evaluation protocol and our cross-spectral generalization task design, we fine-tune all the models on the $C120_{\text{VNIR+}}$ spectral configuration of the training set, selecting models and hyperparameters based on corresponding validation performance, and reporting the results on the test set for all spectral configurations.

Although downstream training and evaluation follow a common protocol, pretraining data and procedures differ across models. SpectralViT is retrained on $C120_{\mathrm{VNIR}+}$, while DOFA, HyperSigma, and HyperFree use their released checkpoints. These differences remain a confounding factor when attributing cross-model performance differences solely to architecture.

D.1 Implementation

As different models and benchmark datasets may have varying optimal learning rates, we conduct a comprehensive search over a wide range of learning rates: {8e-5, 1e-4, 3e-4, 5e-4, 8e-4}. This search is applied to all models and benchmark datasets mentioned above, and we report the best results achieved across all learning rates. During fine-tuning, all models use an effective batch size of 256

and bfloat16 precision. The AdamW Loshchilov and Hutter [2017] optimizer is employed for all models with a weight decay of 1e-2, $\beta_1 = 0.9$, and $\beta_2 = 0.999$. We fine-tune all the models for 20 epochs. Additionally, we utilize a cosine annealing scheduler with a warm-up period lasting 20% of the total fine-tuning epochs to adjust the learning rate during training.

For data augmentation, we start by normalizing all the images. Then we apply random flips both horizontally and vertically, as well as a random rotation. We still avoid any resize operations during data preprocessing, following the data augmentation strategies during pretraining.

E Algorithms

We provide pseudocode for two operations referenced in the main text: full spatial–spectral attention, which serves as a comparison to LESS Attention, and the attention pooling operation used within LESS Attention. The complete LESS Attention block is presented in Algorithm 1.

E.1 Spatial-Spectral Attention Block

Algorithm 2 describes full self-attention over the spatial–spectral token grid, following ChannelViT [Bao et al., 2023]. The $N \times C$ grid is flattened into NC tokens before applying self-attention, resulting in a computational complexity of $\mathcal{O}(N^2C^2D)$.

Algorithm 2 Spatial-Spectral Attention Block Bao et al. [2023]

Require: Input tokens $X \in \mathbb{R}^{N\times C\times D}$
Ensure: Output tokens $Y \in \mathbb{R}^{N\times C\times D}$
$\bar{X} \leftarrow \textsc{Reshape}(X, (NC, D))$ ▷ Flatten input tokens
$Q, K, V \leftarrow \bar{X}W_Q, \bar{X}W_K, \bar{X}W_V \in \mathbb{R}^{NC\times D}$
$Y \leftarrow \textsc{Softmax}(QK^\top/\sqrt{d_k})V \in \mathbb{R}^{NC\times D}$ ▷ Complexity $O(N^2C^2D)$
$Y \leftarrow \textsc{Reshape}(Y, (N, C, D))$
return Y

E.2 Attention Pooling Layer

Algorithm 3 details the attention pooling operation used to construct the spatial and spectral branches of LESS Attention. Along the axis being pooled, the corresponding `[CLS]` token supplies the query, while the input tokens supply the keys and values. The resulting weighted aggregation is projected to the target sub-dimension. Applying this operation along the spectral or spatial axis produces the spatial or spectral branch tokens, respectively.

Algorithm 3 AttenPool

Require: Input tokens $X \in \mathbb{R}^{C\times(N+1)\times D}$, Sub-dimension d
Ensure: Pooled tokens $Y \in \mathbb{R}^{C\times 1\times d}$
Compute query from the `[CLS]` tokens of the dimension to be pooled.
$X_{\texttt{[CLS]}} \leftarrow X[:, :1, :] \in \mathbb{R}^{C\times 1\times D}$ ▷ Extract the attached `[CLS]` token.
$Q \leftarrow X_{\texttt{[CLS]}}W_Q \in \mathbb{R}^{C\times 1\times D}$
Compute key and value from the input tokens.
$K, V \leftarrow XW_K, XW_V \in \mathbb{R}^{C\times(N+1)\times D}$
Use a weighted sum to pool the targeting dimension.
$A \leftarrow \textsc{Softmax}(QK^\top/\sqrt{d_k}) \in \mathbb{R}^{C\times 1\times(N+1)}$ ▷ The weight is essentially the attention map.
$\bar{Y} \leftarrow AV \in \mathbb{R}^{C\times 1\times D}$
$Y \leftarrow \bar{Y}W_d \in \mathbb{R}^{C\times 1\times d}$ ▷ Project to the desired sub-dimension.
return Y

F Additional Experiment Results

F.1 Full Fine-tune and Linear-probing

We report full fine-tuning (FT) and linear probing (LP) results across five downstream datasets and six spectral configurations in Table 1. FT updates the encoder and task head, whereas LP freezes the pretrained encoder and trains only a linear prediction head. Both protocols use $C120_{\mathrm{VNIR+}}$ for downstream training and evaluate the resulting models across all configurations without further adaptation. Results are reported as mean $\pm$ standard deviation across five random seeds.

Under FT, LESSViT achieves the highest mean performance across all 25 out-of-distribution task–configuration pairs. The multi-seed results support the cross-spectral robustness observed in the main experiments, while SpectralViT retains the strongest in-distribution performance. LP further highlights this specialization–robustness trade-off: SpectralViT performs best in distribution but degrades sharply under spectral shifts, whereas LESSViT achieves the highest mean score in 17 of the 25 out-of-distribution pairs. Since LP freezes the encoder, these results indicate that LESSViT's robustness is already encoded in its pretrained representations, rather than arising solely from downstream fine-tuning.

Table 1: **Full fine-tuning (FT) and linear probing (LP) results across five downstream tasks.** Scores are means $\pm$ standard deviations over five seeds; percentages indicate relative drops from the corresponding in-distribution configuration. Bold denotes the highest mean within each dataset–configuration pair under each evaluation protocol.

Model	$C120_{\mathrm{VN+}}$	$C120_{\mathrm{SW+}}$	$C82$	$C202$	PRISMA	EO1H
CDL (mIoU) — Full fine-tuning (FT)						
SpectralViT	**73.33** $\pm$ 1.70	15.11 $\pm$ 1.13 ↓79.4%	12.36 $\pm$ 3.15 ↓83.1%	5.59 $\pm$ 0.17 ↓92.4%	10.84 $\pm$ 2.51 ↓85.2%	7.81 $\pm$ 1.03 ↓89.4%
HyperSigma	44.48 $\pm$ 3.62	13.93 $\pm$ 1.24 ↓68.7%	14.50 $\pm$ 2.91 ↓67.4%	14.51 $\pm$ 0.54 ↓67.4%	10.08 $\pm$ 3.01 ↓77.3%	12.86 $\pm$ 3.02 ↓71.1%
DOFA	40.06 $\pm$ 1.85	18.38 $\pm$ 0.18 ↓54.1%	18.08 $\pm$ 0.75 ↓54.9%	12.39 $\pm$ 0.81 ↓69.1%	13.06 $\pm$ 1.54 ↓67.4%	6.39 $\pm$ 4.10 ↓84.1%
HyperFree	42.50 $\pm$ 0.62	28.48 $\pm$ 1.95 ↓33.0%	16.22 $\pm$ 2.44 ↓61.8%	22.16 $\pm$ 0.70 ↓47.9%	28.09 $\pm$ 1.71 ↓33.9%	19.59 $\pm$ 3.47 ↓53.9%
LESSViT	53.44 $\pm$ 1.02	**48.32** $\pm$ 5.26 ↓9.6%	**42.08** $\pm$ 5.21 ↓21.3%	**50.75** $\pm$ 2.65 ↓5.0%	**48.96** $\pm$ 3.42 ↓8.4%	**49.05** $\pm$ 4.53 ↓8.2%
CDL (mIoU) — Linear probing (LP)						
SpectralViT	**60.65** $\pm$ 0.78	3.75 $\pm$ 0.40 ↓93.8%	3.14 $\pm$ 0.29 ↓94.8%	6.44 $\pm$ 0.23 ↓89.4%	3.71 $\pm$ 0.33 ↓93.9%	3.77 $\pm$ 0.25 ↓93.8%
HyperSigma	27.32 $\pm$ 2.21	18.18 $\pm$ 1.71 ↓33.5%	2.26 $\pm$ 0.51 ↓91.7%	2.09 $\pm$ 0.08 ↓92.3%	6.81 $\pm$ 0.17 ↓75.1%	12.40 $\pm$ 0.82 ↓54.6%
DOFA	31.81 $\pm$ 1.44	4.59 $\pm$ 0.04 ↓85.6%	4.08 $\pm$ 0.20 ↓87.2%	4.14 $\pm$ 0.01 ↓87.0%	6.24 $\pm$ 0.95 ↓80.4%	3.67 $\pm$ 0.53 ↓88.5%
HyperFree	30.12 $\pm$ 0.62	18.99 $\pm$ 0.68 ↓36.9%	15.14 $\pm$ 2.79 ↓49.7%	7.06 $\pm$ 0.23 ↓76.6%	21.46 $\pm$ 0.18 ↓28.8%	22.62 $\pm$ 0.58 ↓24.9%
LESSViT	46.29 $\pm$ 0.51	**31.43** $\pm$ 3.93 ↓32.1%	**30.29** $\pm$ 3.30 ↓34.6%	**39.34** $\pm$ 1.32 ↓15.0%	**29.11** $\pm$ 0.20 ↓37.1%	**31.01** $\pm$ 0.23 ↓33.0%
BDFORET (mIoU) — Full fine-tuning (FT)						
SpectralViT	**77.17** $\pm$ 0.50	48.31 $\pm$ 0.46 ↓37.4%	48.68 $\pm$ 0.45 ↓36.9%	48.73 $\pm$ 0.21 ↓36.9%	46.77 $\pm$ 1.04 ↓39.4%	44.55 $\pm$ 2.40 ↓42.3%
HyperSigma	63.76 $\pm$ 0.33	47.97 $\pm$ 1.32 ↓24.8%	44.15 $\pm$ 3.64 ↓30.8%	48.94 $\pm$ 1.13 ↓23.2%	43.68 $\pm$ 2.62 ↓31.5%	9.54 $\pm$ 2.19 ↓85.0%
DOFA	58.14 $\pm$ 0.50	51.26 $\pm$ 0.63 ↓11.8%	42.03 $\pm$ 4.75 ↓27.7%	50.45 $\pm$ 0.19 ↓13.2%	45.04 $\pm$ 3.44 ↓22.5%	2.36 $\pm$ 1.37 ↓95.9%
HyperFree	58.42 $\pm$ 2.09	54.43 $\pm$ 2.85 ↓6.8%	52.63 $\pm$ 3.02 ↓9.9%	54.13 $\pm$ 1.59 ↓7.3%	54.44 $\pm$ 1.55 ↓6.8%	53.93 $\pm$ 1.60 ↓7.7%
LESSViT	63.87 $\pm$ 0.61	**62.43** $\pm$ 2.02 ↓2.3%	**60.58** $\pm$ 3.68 ↓5.2%	**63.35** $\pm$ 0.54 ↓0.8%	**62.03** $\pm$ 0.83 ↓2.9%	**62.60** $\pm$ 0.55 ↓2.0%
BDFORET (mIoU) — Linear probing (LP)						
SpectralViT	**66.58** $\pm$ 0.25	46.70 $\pm$ 0.08 ↓29.9%	46.23 $\pm$ 0.10 ↓30.6%	33.75 $\pm$ 0.31 ↓49.3%	47.37 $\pm$ 0.06 ↓28.9%	41.60 $\pm$ 1.90 ↓37.5%
HyperSigma	50.75 $\pm$ 0.44	45.58 $\pm$ 3.03 ↓10.2%	6.67 $\pm$ 0.00 ↓86.9%	46.71 $\pm$ 3.25 ↓8.0%	21.54 $\pm$ 2.90 ↓57.6%	48.80 $\pm$ 0.00 ↓3.8%
DOFA	55.75 $\pm$ 0.76	4.26 $\pm$ 0.03 ↓92.4%	4.19 $\pm$ 0.08 ↓92.5%	3.88 $\pm$ 0.51 ↓93.0%	3.62 $\pm$ 0.03 ↓93.5%	3.55 $\pm$ 0.06 ↓93.6%
HyperFree	53.84 $\pm$ 0.30	**49.08** $\pm$ 1.34 ↓8.8%	50.11 $\pm$ 0.68 ↓6.9%	49.86 $\pm$ 1.56 ↓7.4%	**50.10** $\pm$ 0.05 ↓6.9%	**49.10** $\pm$ 0.08 ↓8.8%
LESSViT	59.60 $\pm$ 0.02	42.31 $\pm$ 0.71 ↓29.0%	**51.35** $\pm$ 2.56 ↓13.9%	**53.26** $\pm$ 0.23 ↓10.6%	45.85 $\pm$ 0.33 ↓23.1%	45.76 $\pm$ 0.36 ↓23.2%
BNETD (mIoU) — Full fine-tuning (FT)						

Continued on next page

Table 1 (continued)

Model	$C120_{\mathrm{VN}+}$	$C120_{\mathrm{SW}+}$	$C82$	$C202$	PRISMA	EO1H
SpectralViT	$\mathbf{44.69} \pm 0.64$	16.86 ± 2.59 ↓62.3%	17.06 ± 1.01 ↓61.8%	16.66 ± 1.33 ↓62.7%	18.00 ± 1.42 ↓59.7%	16.07 ± 3.03 ↓64.1%
HyperSigma	30.36 ± 2.96	18.78 ± 0.90 ↓38.1%	17.64 ± 0.73 ↓41.9%	17.68 ± 1.98 ↓41.8%	15.31 ± 0.17 ↓49.6%	7.22 ± 0.38 ↓76.2%
DOFA	28.88 ± 2.20	18.06 ± 0.04 ↓37.5%	18.87 ± 0.30 ↓34.7%	18.89 ± 0.23 ↓34.6%	15.75 ± 0.18 ↓45.5%	6.29 ± 0.48 ↓78.2%
HyperFree	29.53 ± 0.86	18.53 ± 2.19 ↓37.3%	16.87 ± 1.93 ↓42.9%	21.47 ± 1.98 ↓27.3%	21.93 ± 0.53 ↓25.7%	18.97 ± 0.28 ↓35.8%
LESSViT	32.37 ± 0.82	$\mathbf{30.76} \pm 2.57$ ↓5.0%	$\mathbf{26.35} \pm 4.28$ ↓18.6%	$\mathbf{31.25} \pm 1.85$ ↓3.5%	$\mathbf{31.80} \pm 0.30$ ↓1.8%	$\mathbf{29.62} \pm 3.07$ ↓8.5%
			BNETD (mIoU) — Linear probing (LP)			
SpectralViT	$\mathbf{37.55} \pm 0.63$	10.40 ± 0.39 ↓72.3%	11.38 ± 0.39 ↓69.7%	10.67 ± 0.51 ↓71.6%	10.41 ± 0.08 ↓72.3%	9.55 ± 0.04 ↓74.6%
HyperSigma	21.35 ± 0.71	12.58 ± 0.50 ↓41.1%	5.41 ± 0.01 ↓74.7%	8.62 ± 5.36 ↓59.6%	5.22 ± 1.05 ↓75.5%	8.04 ± 0.42 ↓62.4%
DOFA	24.54 ± 0.40	8.44 ± 1.31 ↓65.6%	5.10 ± 0.49 ↓79.2%	14.80 ± 0.68 ↓39.7%	8.27 ± 0.81 ↓66.3%	8.10 ± 1.16 ↓67.0%
HyperFree	24.84 ± 0.54	15.64 ± 0.65 ↓37.0%	$\mathbf{16.76} \pm 2.33$ ↓32.5%	12.71 ± 0.03 ↓48.8%	11.75 ± 0.67 ↓52.7%	$\mathbf{19.59} \pm 0.32$ ↓21.2%
LESSViT	30.11 ± 0.08	$\mathbf{17.73} \pm 0.49$ ↓41.1%	14.42 ± 0.40 ↓52.1%	$\mathbf{23.05} \pm 0.52$ ↓23.4%	$\mathbf{17.00} \pm 1.07$ ↓43.5%	18.20 ± 1.40 ↓39.6%
			EuroCrops (mIoU) — Full fine-tuning (FT)			
SpectralViT	$\mathbf{71.69} \pm 0.79$	26.27 ± 5.33 ↓63.4%	41.14 ± 1.13 ↓42.6%	46.28 ± 1.07 ↓35.4%	42.23 ± 2.19 ↓41.1%	29.32 ± 10.24 ↓59.1%
HyperSigma	56.54 ± 0.28	52.16 ± 1.17 ↓7.7%	48.45 ± 3.36 ↓14.3%	51.59 ± 0.14 ↓8.8%	46.32 ± 2.57 ↓18.1%	38.84 ± 2.96 ↓31.3%
DOFA	55.59 ± 0.63	36.85 ± 5.04 ↓33.7%	32.83 ± 1.94 ↓40.9%	31.52 ± 3.17 ↓43.3%	50.28 ± 0.27 ↓9.5%	50.40 ± 0.004 ↓9.3%
HyperFree	52.22 ± 2.10	46.63 ± 1.09 ↓10.7%	32.87 ± 6.30 ↓37.1%	36.25 ± 7.10 ↓30.6%	45.71 ± 2.00 ↓12.5%	20.91 ± 6.14 ↓60.0%
LESSViT	57.91 ± 0.61	$\mathbf{55.69} \pm 5.11$ ↓3.8%	$\mathbf{55.47} \pm 2.48$ ↓4.2%	$\mathbf{57.81} \pm 1.64$ ↓0.2%	$\mathbf{57.63} \pm 1.42$ ↓0.5%	$\mathbf{57.48} \pm 1.17$ ↓0.7%
			EuroCrops (mIoU) — Linear probing (LP)			
SpectralViT	$\mathbf{65.62} \pm 0.07$	28.27 ± 0.10 ↓56.9%	12.30 ± 0.11 ↓81.3%	31.87 ± 0.08 ↓51.4%	31.27 ± 1.69 ↓52.3%	$\mathbf{46.43} \pm 0.33$ ↓29.2%
HyperSigma	51.13 ± 0.01	$\mathbf{50.43} \pm 0.28$ ↓1.4%	23.06 ± 2.71 ↓54.9%	27.53 ± 0.20 ↓46.2%	34.08 ± 1.14 ↓33.3%	43.16 ± 0.45 ↓15.6%
DOFA	53.70 ± 0.34	30.11 ± 5.26 ↓43.9%	16.80 ± 3.03 ↓68.7%	38.76 ± 2.42 ↓27.8%	$\mathbf{47.08} \pm 2.98$ ↓12.3%	43.77 ± 3.44 ↓18.5%
HyperFree	51.04 ± 0.03	33.68 ± 3.69 ↓34.0%	29.06 ± 3.82 ↓43.1%	17.76 ± 3.21 ↓65.2%	43.69 ± 3.66 ↓14.4%	35.89 ± 4.66 ↓29.7%
LESSViT	54.58 ± 3.73	45.29 ± 4.55 ↓17.0%	$\mathbf{44.11} \pm 5.32$ ↓19.2%	$\mathbf{51.44} \pm 1.81$ ↓5.8%	45.23 ± 1.42 ↓17.1%	45.86 ± 3.50 ↓16.0%
			CORINE (mAP) — Full fine-tuning (FT)			
SpectralViT	$\mathbf{78.93} \pm 0.19$	36.90 ± 1.28 ↓53.2%	36.92 ± 2.15 ↓53.2%	38.37 ± 1.07 ↓51.4%	35.24 ± 2.59 ↓55.4%	33.08 ± 1.81 ↓58.1%
HyperSigma	69.21 ± 0.52	36.05 ± 4.65 ↓47.9%	27.56 ± 1.06 ↓60.2%	47.11 ± 2.57 ↓31.9%	38.60 ± 4.03 ↓44.2%	39.48 ± 0.91 ↓43.0%
DOFA	66.40 ± 6.96	37.89 ± 2.99 ↓42.9%	48.27 ± 2.94 ↓27.3%	35.33 ± 1.15 ↓46.8%	25.66 ± 7.74 ↓61.4%	17.58 ± 0.19 ↓73.5%
HyperFree	74.04 ± 0.76	47.93 ± 2.67 ↓35.3%	63.31 ± 4.67 ↓14.5%	68.48 ± 0.22 ↓7.5%	65.83 ± 0.59 ↓11.1%	62.28 ± 1.50 ↓15.9%
LESSViT	76.35 ± 0.46	$\mathbf{74.21} \pm 3.55$ ↓2.8%	$\mathbf{72.95} \pm 4.05$ ↓4.5%	$\mathbf{75.99} \pm 0.30$ ↓0.5%	$\mathbf{75.05} \pm 0.82$ ↓1.7%	$\mathbf{75.01} \pm 1.43$ ↓1.8%
			CORINE (mAP) — Linear probing (LP)			
SpectralViT	$\mathbf{71.33} \pm 0.21$	37.55 ± 0.06 ↓47.4%	39.40 ± 0.07 ↓44.8%	37.05 ± 0.47 ↓48.1%	49.42 ± 0.10 ↓30.7%	34.64 ± 0.08 ↓51.4%
HyperSigma	61.66 ± 0.14	41.76 ± 1.65 ↓32.3%	24.15 ± 1.10 ↓60.8%	35.01 ± 0.39 ↓43.2%	42.79 ± 0.04 ↓30.6%	40.93 ± 1.91 ↓33.6%
DOFA	52.18 ± 1.16	19.72 ± 0.61 ↓62.2%	11.05 ± 0.52 ↓78.8%	30.05 ± 0.71 ↓42.4%	30.88 ± 0.68 ↓40.8%	29.64 ± 0.17 ↓43.2%
HyperFree	63.11 ± 0.08	49.32 ± 0.03 ↓21.8%	45.24 ± 1.14 ↓28.3%	43.61 ± 0.50 ↓30.9%	46.10 ± 1.07 ↓27.0%	50.31 ± 0.08 ↓20.3%
LESSViT	69.96 ± 0.02	$\mathbf{60.26} \pm 1.07$ ↓13.9%	$\mathbf{57.67} \pm 0.91$ ↓17.6%	$\mathbf{63.97} \pm 0.96$ ↓8.6%	$\mathbf{61.67} \pm 0.69$ ↓11.9%	$\mathbf{60.56} \pm 0.88$ ↓13.4%

F.2 Effective Rank of Pretrained Representations

To examine whether the low-rank structure of LESS Attention limits the dimensionality of learned representations, we measure the effective rank of pretrained encoder features using RankMe [Garrido et al., 2023], normalized by the embedding dimension ($D = 768$). HyperSigma is excluded because its dual branches are fused only at the task head and do not provide a unified intermediate representation for this comparison.

As shown in Table 2, LESSViT exhibits the highest normalized effective rank across all four spectral configurations. These results indicate that its structured low-rank attention does not lead to low-rank encoder features. Instead, LESSViT learns representations that occupy a larger effective fraction of the embedding space than the compared models.

Table 2: **Effective rank across spectral configurations.** Bold indicates the highest value in each configuration.

Model	$C120_{\mathrm{VNIR+}}$	$C120_{\mathrm{SWIR+}}$	$C82$	$C202$
SpectralViT	0.255	0.275	0.259	0.275
DOFA	0.092	0.087	0.083	0.091
HyperFree	0.260	0.244	0.246	0.245
LESSViT	**0.343**	**0.362**	**0.361**	**0.358**

F.3 In-Distribution Performance and Cross-Spectral Robustness

To quantify the trade-off between in-distribution specialization and cross-spectral robustness, we report ID performance, average OOD performance, and retention rates in Table 3, computed from multi-seed fine-tuning means. ID denotes $C120_{\mathrm{VNIR+}}$, while OOD averages over $C120_{\mathrm{SWIR+}}$, $C82$, $C202$, and the simulated PRISMA and Hyperion configurations. Segmentation mIoU is averaged across CDL, BDFORET, BNETD, and EuroCrops; classification mAP is reported on CORINE. Retention is the ratio of aggregated OOD to ID performance, expressed as a percentage.

SpectralViT achieves the strongest in-distribution performance but retains only 41.86% of its segmentation mIoU and 45.74% of its classification mAP under spectral shifts. In contrast, LESSViT retains 94.80% and 97.76%, respectively, while achieving the highest average OOD performance for both metrics. These results highlight the trade-off between fixed-configuration specialization and cross-spectral robustness, with LESSViT providing a more favorable balance.

Table 3: **In-distribution performance and out-of-distribution robustness.** Bold indicates the best result in each column.

	Segmentation (mIoU)			Classification (mAP)		
Model	ID	Avg. OOD	Retention (%)	ID	Avg. OOD	Retention (%)
SpectralViT	**66.72**	27.93	41.86	**78.93**	36.10	45.74
HyperSigma	48.78	28.71	58.84	69.21	37.76	54.56
DOFA	45.67	26.96	59.03	66.40	32.95	49.62
HyperFree	45.67	33.21	72.73	74.04	61.57	83.15
LESSViT	51.90	**49.20**	**94.80**	76.35	**74.64**	**97.76**

F.4 Attention and Component Ablations

Attention operators. We provide the per-dataset results of the controlled attention comparison in Table 4. All three models use a ViT-S backbone with 64×64 inputs and identical HyperMAE pretraining, including decoupled masking, hierarchical channel sampling, training data, and optimization settings. Spatial-only attention achieves strong in-distribution performance but degrades sharply under spectral reconfiguration, showing that broader spectral exposure during pretraining alone does not explain the robustness gains. LESS and full spatial–spectral attention achieve broadly comparable performance, with neither consistently outperforming the other, supporting LESS as a scalable alternative to full attention.

Positional encoding and channel sampling. Table 5 reports the full component-ablation results. SSRoPE improves average segmentation performance over SIREN across all spectral configurations. Increasing the HCS retained ratio r_{HCS} from $[0.2, 0.3]$ to $[0.4, 0.5]$ yields modest changes in downstream performance, while the lower ratio reduces pretraining time from 41 to 34 hours. Tied patch embedding and decoupled masking are structural requirements of the proposed formulation: the former supports varying channel counts, while the latter preserves the axis-aligned token grid required by the factorized attention.

Table 4: **Comparison of attention operators under spectral configuration shift.** Results are reported as mIoU for segmentation and mAP for CORINE classification. Bold indicates the best score in each column.

	CDL (mIoU)				EuroCrops (mIoU)				CORINE (mAP)			
Attention operator	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$
Spatial Attention	**58.92**	7.41	3.20	15.52	**62.12**	42.11	30.79	49.71	**76.97**	29.32	26.37	33.51
Full Spatial–Spectral Attention	45.04	41.95	42.64	37.66	57.84	**57.40**	**57.34**	**57.44**	71.25	70.17	70.78	69.47
LESS Attention	44.22	**42.44**	**42.85**	**40.47**	57.72	56.15	56.10	55.73	73.19	**71.18**	**72.14**	**70.00**

	BDFORET (mIoU)				BNETD (mIoU)			
Attention operator	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$
Spatial Attention	**68.26**	32.12	21.87	48.10	**35.87**	9.39	8.46	3.68
Full Spatial–Spectral Attention	62.81	**58.85**	58.53	**58.25**	30.51	**28.61**	**27.45**	**26.11**
LESS Attention	63.51	57.75	**59.92**	56.60	31.29	27.95	17.01	18.72

Table 5: **Ablation of key modules under cross-spectral generalization.** Models are trained on $C120_{\text{VN+}}$ and evaluated across spectral configurations ($C120_{\text{VN+}}$, $C120_{\text{SW+}}$, $C82$, $C202$). We compare positional encoding (SSRoPE vs. SIREN) and hierarchical channel sampling (HCS) ratios. Metrics include mIoU for segmentation tasks and mAP for classification tasks.

			CDL (mIoU)				EuroCrops (mIoU)				CORINE (mAP)			
Config.	PE	r_{HCS}	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$
Default	SSRoPE	$[0.2, 0.3]$	40.04	41.22	35.60	40.50	52.52	46.94	47.49	49.78	70.74	68.30	67.11	69.61
w/o SSRoPE	SIREN	$[0.2, 0.3]$	28.79	29.60	30.58	29.82	50.45	45.77	46.01	47.27	70.32	69.48	64.68	69.89
High r_{HCS}	SSRoPE	$[0.4, 0.5]$	40.93	40.87	35.88	39.13	52.06	51.54	48.70	51.37	69.68	69.04	64.61	69.83

BDFORET (mIoU)				BNETD (mIoU)			
$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$	$C120_{\text{VN+}}$	$C120_{\text{SW+}}$	$C82$	$C202$
57.30	58.22	54.39	58.11	28.04	24.24	22.37	26.14
55.51	54.14	53.89	54.41	25.41	21.51	20.89	23.08
59.81	58.19	54.45	58.85	28.95	25.77	23.76	26.44

F.5 Computational Efficiency

We report fine-tuning and inference costs in Table 6, measured on CORINE in BF16 with effective batch sizes of 32 and 1, respectively. Since ChannelViT exceeds GPU memory at a fine-tuning batch size of 32, its step time is measured using a microbatch size of 4 with eight gradient accumulation steps per optimizer update.

Compared with ChannelViT, LESSViT achieves approximately a 29-fold reduction in fine-tuning step time, a 7-fold reduction in inference latency, and an 11-fold reduction in inference memory, while requiring approximately 2.6 times less estimated peak fine-tuning memory. It remains more expensive to train than channel-collapsing methods because it retains the explicit $N \times C$ spatial–spectral token grid. Our efficiency advantage is therefore relative to full spatial–spectral attention. Profiling shows that LESS Attention accounts for only 1.17 GB, approximately 1.3% of peak training memory, whereas MLP activations account for approximately 43%. The remaining memory overhead thus primarily reflects processing the explicit token grid rather than constructing a quadratic attention matrix. Pretraining costs are reported in Appendix C.

Table 6: **Computational efficiency.** Encoder parameter counts and fine-tuning and inference costs.

	Encoder	Fine-tuning		Inference	
Model	params (M)	Step time (s)	Peak memory (GB)	Latency (ms)	Peak memory (GB)
SpectralViT-B	87.5	0.39	14.9	7.1	0.64
HyperSigma-B	203.0	0.17	12.2	32.0	1.31
DOFA-B	111.8	0.035	1.93	13.3	0.87
HyperFree-B	151.5	0.18	2.25	52.8	0.94
ChannelViT-B*	85.5	21.0	∼230	221.4	7.73
LESSViT-B	83.2	0.73	87.2	30.5	0.67

*ChannelViT exceeds GPU memory at batch size 32; the approximately 230 GB memory requirement is estimated from the OOM error. Its step time is measured using a microbatch size of 4 with eight gradient accumulation steps per optimizer update.

F.6 Additional Qualitative Results

We provide additional qualitative results for segmentation tasks in Figure 6.



Figure 6: **Additional qualitative results for segmentation tasks.** We visualize the segmentation maps generated by SpectralViT, LESSViT, HyperSigma, and DOFA on different tasks and channel configurations.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction clearly state the contributions of LESSViT, including its scalable spatial–spectral attention architecture and cross-spectral generalization capabilities. These claims are supported by the architectural analysis, downstream evaluations, controlled ablations, and computational efficiency experiments.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The conclusion explicitly discusses several limitations, including the use of EnMAP-derived spectral configurations rather than real cross-sensor observations, untested differences in sensor calibration and imaging conditions, increasing computational demands at higher spatial and spectral resolutions, and the lack of validation beyond geospatial hyperspectral imagery.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: This work introduces an attention architecture supported by mathematical formulations and computational complexity analysis, but does not establish formal theoretical guarantees or present theorems requiring proofs.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The paper and appendix describe the proposed architecture, pretraining and fine-tuning procedures, spectral configurations, evaluation protocols, and baseline comparisons. These details, together with the released implementation, support reproduction of the main experimental results.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The implementation of LESSViT is publicly available through the accompanying GitHub repository. The experiments use publicly accessible remote sensing datasets, and the paper and supplementary material describe the data configurations and evaluation procedures.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The experimental sections and appendix document the datasets, spectral band configurations, pretraining and fine-tuning protocols, model configurations, optimization settings, and evaluation metrics. These details specify how the reported experiments and cross-spectral evaluations are conducted.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: We report mean performance and standard deviation across five random seeds for the main fine-tuning and linear-probing evaluations. These statistics characterize run-to-run variability under fixed experimental settings and are provided in Appendix F.1.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The paper and appendix report the computational resources and costs associated with model training and evaluation, including GPU memory consumption, training time, fine-tuning step time, and inference latency. We additionally analyze the computational scaling of the proposed attention architecture.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research follows the NeurIPS Code of Ethics. The experiments use existing remote sensing datasets and computational models, without involving human subjects or collecting personal information.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [No]

    Justification: This work primarily focuses on the architectural design and computational efficiency of hyperspectral representation learning. While potential applications include environmental monitoring and land-cover analysis, broader societal impacts and potential risks are not explicitly examined in this paper.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The proposed method focuses on hyperspectral remote sensing representation learning and does not involve the release of high-risk generative models, sensitive personal data, or other assets requiring specialized safeguards.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All existing datasets and pretrained models used in this work are properly attributed to their original sources through appropriate citations. We use publicly accessible remote sensing datasets and respect their applicable licenses and terms of use.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The proposed method is accompanied by publicly available source code and implementation details. The paper and supplementary material document the model architecture, training configurations, evaluation protocols, and instructions needed to reproduce the reported results.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: This research does not involve crowdsourcing or human subjects. All experiments are conducted using remote sensing datasets and computational models.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: This research does not involve human participants, human subject data collection, or interventions involving human subjects. Therefore, IRB approval is not applicable.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: Large language models are not used as a component of the proposed methodology, model training, or experimental evaluation.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.